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Synergistic Mechanisms for Enhanced Waterflooding Development in Middle Eastern Thin-Bedded Carbonate Reservoirs

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Abstract

Middle Eastern thin-bedded carbonate reservoirs exhibit good lateral connectivity but suffer from extreme vertical heterogeneity. High-permeability streaks at the reservoir top act as "thief zones," leading to severe sweep bypass and low recovery factors. These features pose significant challenges for conventional waterflooding. Using the geological parameters of the M reservoir (A Oilfield), we constructed a large-scale 3D artificial core model (60×60×6 cm^3) to evaluate three sub-layers. We compared three strategies: continuous stable waterflooding, unstable waterflooding with water shutoff, and unstable waterflooding with sidetracking.

Experiments showed that while stable waterflooding achieved only a 27.3% recovery factor, combining unstable waterflooding with water shutoff or sidetracking increased recovery to 32.8% and 35.7%, respectively. The key mechanisms include: (1) alternating unstable waterflooding activates reservoir elasticity, doubling the inter-well dimensionless pressure gradient (DPG) and expanding sweep efficiency by 12%–18%; (2) water shutoff reconfigures flow paths by plugging high-permeability channels; and (3) sidetracking establishes new flow channels in low-permeability layers. The "unstable waterflooding combined with sidetracking" approach demonstrated the strongest synergistic effect, reaching oil utilization efficiencies of 43.7% in high-perm layers and 29.5% in low-perm layers. This study provides a theoretical framework for enhancing recovery in heterogeneous thin-bedded reservoirs.

Introduction

As fossil fuels are projected to remain the cornerstone of the global energy mix for the foreseeable future, carbonate reservoirs have become indispensable to global energy security (Brockway et al. 2019, Chen et al. 2025, Song et al. 2026). These reservoirs harbor approximately 70% of the world's oil reserves and account for 60% of total production, underscoring their critical role in meeting global energy demand (Lucia et al. 2003, Li et al. 2018, Mogensen and Masalmeh 2020). Waterflooding, as the most widely applied method for replenishing reservoir energy and displacing crude oil post primary recovery, plays an especially critical role in such reservoirs (Cao et al. 2024, Cao et al. 2025, Wang et al. 2025). The strongly heterogeneous, thin-bedded carbonate reservoirs widely distributed in the Middle East present formidable barriers to sustained production growth under waterflooding. Premature water breakthrough, accelerated decline rates, elevated injection pressures, and difficulty in mobilizing remaining oil (Chen et al. 2020, Xu et al. 2020, Li et al. 2025), which are direct consequences of the reservoirs' inherent features: strong vertical heterogeneity, extensive interlayer development, and significant contrasts in petrophysical properties across different layers (Song and Li 2018, Xu et al. 2020, Cross and Burchette 2025).

To improve waterflooding performance and production growth in carbonate reservoirs, various field trials have been conducted worldwide. These include water-based techniques like unstable waterflooding (Li et al. 2021), and chemical-based methods such as polymer flooding and deep profile control/water shutoff (Diab and Al-Shalabi 2019, Alabdrabalnabi et al. 2023, Al-Mayyan et al. 2024), all of which have demonstrated notable success. For instance, unstable water injection in the North Truva field resulted in a cumulative incremental oil production of 12,048 tons in the pilot area, with an average daily incremental output of 3.6 tons per well (Li et al. 2023). In the Ahdeb oilfield, alternating unstable waterflooding led to a maximum production increase of about 30% and an initial water cut reduction exceeding 15%. Furthermore, a pilot test in the target reservoir integrating unstable alternating injection-production with horizontal well sidetracking and water shutoff technology showed promising results (Hu et al. 2023). Within a test area involving 9 injectors and 6 producers, the maximum water cut reduction reached 19.8%, with approximately 1.7 million barrels of cumulative incremental oil. While field trials of enhanced waterflooding developments have shown potential, their underlying mechanisms and optimal application strategies to achieve sustained production growth remain poorly understood.

Several studies have investigated the mechanisms of unstable waterflooding using both experimental and numerical approaches. Zhou et al. used parallel dual-core experiments to demonstrate that pressure fluctuations overcome the Jamin effect, thereby expanding the swept volume, while capillary forces promote fluid exchange during the injection reduction phase (Zhou et al. 2022). Zhao et al. further indicated that this method promotes interlayer crossflow, which improves vertical sweep efficiency (Zhao et al. 2025). Similarly, numerical simulations by Yang et al. showed that unstable waterflooding alters streamline directions to reach previously unswept oil (Yang et al. 2021). To address remaining oil mobilization during high water-cut stages, Ge et al. verified that sidetracking existing wells effectively rebalances drive energy from gas caps and edge water, significantly reducing oil saturation (Ge et al. 2023). However, most existing research focuses on the recovery mechanisms of individual measures. A notable gap remains in the systematic physical simulation and quantitative analysis of the dynamic succession and spatial synergy between multiple strategies across different development stages, particularly during the ultra-high water-cut period.

Addressing this knowledge gap is essential for developing effective strategies to sustain production growth in maturing Middle Eastern carbonate reservoirs. To this end, we employ a large-scale 3D physical simulation platform configured with a single-injector/producer well pattern, constructing a three-layer heterogeneous model that captures the key geological features of thin-bedded carbonate reservoirs. The full development lifecycle is simulated—progressing from continuous waterflooding (CWF) to unstable waterflooding, followed by either water shutoff or sidetracking—with real-time monitoring of pressure and saturation field dynamics throughout each phase. By tracking reservoir responses at key field-scale decision points—specifically at water cut (f_w) of 70% (transition to unstable flooding), 85% (trigger for well interventions), and 98% (economic limit). This study aims to: (1) elucidate the synergistic mechanisms by which unstable waterflooding and flow-field adjustment strategies (sidetracking and water shutoff) drive incremental production, and (2) establish the layer-specific production contribution sequence across the entire reservoir lifecycle to guide field-scale decision-making. The findings provide both a theoretical foundation and practical guidance for maximizing ultimate recovery and precisely mobilizing remaining reserves in thin-bedded carbonate reservoirs during high water-cut stages.

Experimental section

Materials

Carbonate models. Based on the geological parameters of the M Oilfield (K reservoir), a large-scale three-dimensional heterogeneous carbonate model (60 × 60 × 6 cm^3) was constructed, featuring three representative layers: high-, mid-, and low-permeability. The model integrates internal saturation probes and pressure sensors, operating at conditions up to 80°C and 2 MPa. Two models for three comparative experiments were designed to evaluate recovery strategies across dimensions of injection-production strategy, well pattern adjustment, and system optimization (see Fig. 1 for design of physical models): (a) continuous stable waterflooding; (b) unstable waterflooding combined with water shutoff; and (c) unstable waterflooding combined with sidetracking.

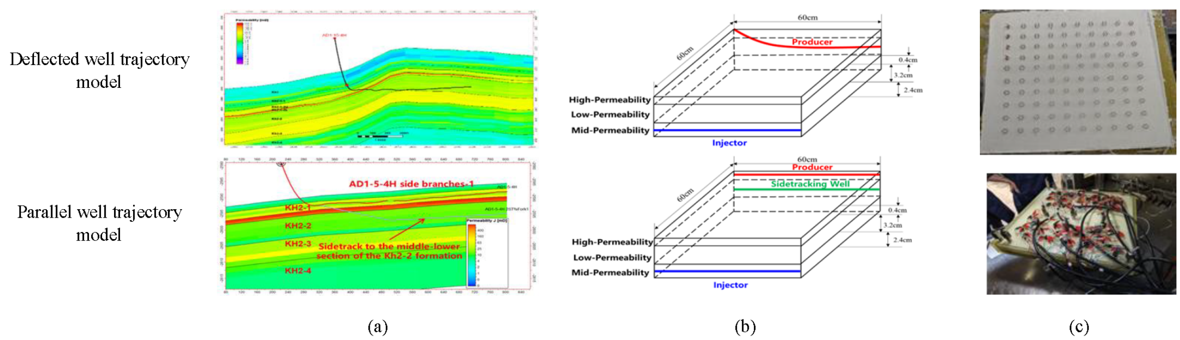

Figure 1—Design of three-dimensional physical models. (a) Oilfield models; (b) Designed models; (c) Physical models

Scheme (a) and (b) – Deflected well pattern: These schemes utilize a central injection well in the mid-perm layer and a Deflected production well penetrating both high- and low-perm layers (simulating a deflected well trajectory). While Scheme (a) serves as a continuous stable waterflooding baseline, Scheme (b) incorporates unstable waterflooding once water cut reaches 70% and a water shutoff operation (60 mL gel injection via the producer followed by a 3-day gel setting period) once water cut reaches 85%.

Scheme (c) – Sidetracking pattern: This scheme employs a parallel well trajectory. Initially, only the production wellbore in the high-perm layer is active. Upon reaching the 85% water-cut threshold, the production is switched to a new sidetracked wellbore completed exclusively in the low-perm layer to bypass the "thief zones."

The specific operational sequences and parameters for all schemes are summarized in Table 1. The key similarity parameters, referencing field horizontal well patterns and production data, are summarized in Table 2.

Table 1—Specific operational sequences and parameters for different schemes.

Models	Strategies	Experimental schemes
Deflected well trajectory model	Continuous stable waterflooding	Continuous waterflooding until the water cut reached 98%.
Deflected well trajectory model	Unstable waterflooding + Water shutoff	Stable waterflooding to a 70% water cut, followed by unstable waterflooding until the water cut reached 85%. After water shutoff was conducted in the production well, stable waterflooding continued until the final water cut reached 98%.
Parallel well trajectory model	Unstable waterflooding + Sidetracking	Stable waterflooding to a 70% water cut, followed by unstable waterflooding until the water cut reached 85%. After sidetracking was conducted in the production well, stable waterflooding continued until the final water cut reached 98%.

Table 2—Design of experimental similarity parameter.

Parameters	Oilfield geological values	Laboratory experimental values
Porosity	19%	19%
Permeability	500/30/10 mD	530/32/11 mD
Inter-well distance	300 m	0.06 m
Horizontal well length	300 m	0.06 m
Thickness (High-/Low-/Mid-perm layer)	0.4/3.2/2.4 cm	1/8/6m
Injection rate	53 m^3/d	2 ml/min
Alternating unstable waterflooding cycle	60 d	40 min

Fluid. The experimental oil had a density of 912.3 kg/m^3 (20°C, 0.101 MPa) and a viscosity of 33.1 mPa·s (80°C). To simulate the flow characteristics under reservoir conditions (viscosity of 1.5 mPa·s), degassed crude oil was mixed with kerosene in a specific ratio and injected into the model. Simulated formation water was prepared based on the ionic composition of the field injection water (composition shown in Table 3), and the water shutoff agent was the same as that used in the field.

Table 3—The ionic composition of the artificial brine water used in the experiments.

Salinity(mg/L)	The ionic composition (mg/L)						
183,260	CO_3^{2-}	HCO_3^-	Cl^-	SO_4^{2-}	Ca^{2+}	Mg^{2+}	K^++Na^+
	0	399	111,221	700	7,461	453	63,226

Experimental set-up

The experimental setup mainly consists of an injection/production fluid system, a measurement system, a data acquisition and monitoring system, and the three-dimensional physical model (see Fig. 2). The injection system uses ISCO pumps and piston containers; the production system includes flow controllers, suction pumps, and intermediate containers to achieve unstable waterflooding. The measurement system comprises gas-liquid separators and gas-liquid meters. Pressure monitoring was achieved using differential pressure transmitters (strategically positioned along the mid-sections of the model) coupled with an M400 data acquisition system. Concurrently, saturation monitoring was performed by measuring resistivity changes via a multimeter, enabling the inversion of both waterflood front advancement and saturation field distribution.

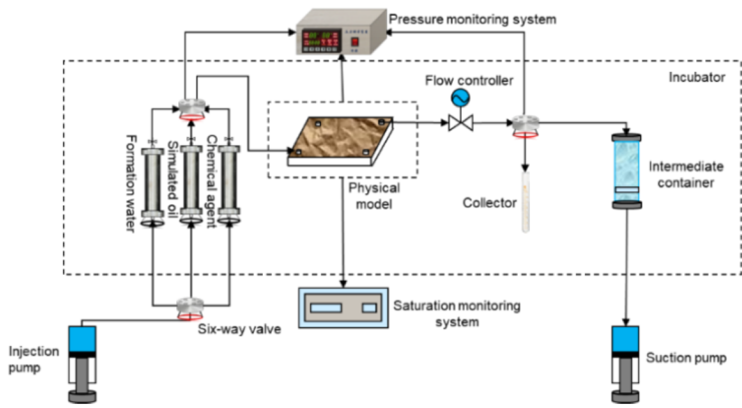

Figure 2—Schematic of experimental set-up for unstable waterflooding

Experimental procedure

The experimental process is comprised of the following five primary stages:

1. Model Integrity Check: The hermetic seal and airtightness of the physical model are rigorously tested under experimental pressure conditions to ensure leakage-free operation throughout the process.
2. Saturation and Reservoir Aging: The model is first evacuated for 18 hours using a vacuum pump, followed by brine saturation. Crude oil is then injected to displace the brine, establishing the connate water saturation (S_{wi}) and initial oil saturation. To restore reservoir-representative wettability, the model is subsequently aged in a constant-temperature oven at 80°C for 7 days.
3. Baseline Waterflooding: The experimental setup is connected, and water injection is initiated at a constant flow rate of 2 mL/min. In the control group (continuous waterflooding), injection is maintained until the terminal water cut reaches 98%. For all enhanced waterflooding development schemes, the baseline waterflooding is halted once the water cut reaches a 70% threshold.
4. Implementation of Improved Waterflooding Measures: Based on the specific experimental design, the following enhanced waterflooding procedures are implemented:

    a. Cyclic Unstable Waterflooding: For this designated schemes, cyclic injection-production is implemented with a 40-minute cycle duration. Each 40-minute cycle comprised two distinct phases: (i) high-injection/low-production phase (IPR = 1.2:0.8), where the injection rate was prioritized over the production rate; and (ii) low-injection/high-production phase (IPR = 0.8:1.2), where the injection rate is reduced and production is intensified. This cyclic operation continues until the composite water cut reaches 85%.
    b. Post-Treatment Waterflooding: Following the execution of either water shutoff or sidetracking measures, subsequent stable waterflooding (post-shutoff/sidetracking waterflooding) is resumed and continued until a terminal water cut of 98% is achieved.
5. Data Acquisition and Monitoring: Throughout the experimental duration, key parameters—including produced fluid volume, resistivity values (for saturation monitoring), displacement pressure, and oil recovery factor—are continuously recorded as functions of the cumulative pore volume (PV) of injected fluid. Fig. 3 illustrates the schematic of experimental procedures for different schemes.

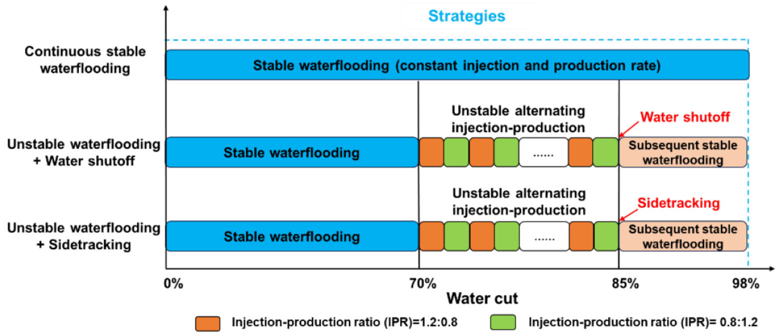

Figure 3—Schematic of the experimental procedure

Mechanisms of Synergistic EOR: From Interlayer Interference Mitigation to Low-Permeability Layer Activation

Overview: Development Stage Transitions and Production Growth Bottlenecks

To assess how vertical heterogeneity affects development performance, we first characterize the dynamics of continuous waterflooding (CWF) using a three-dimension physical simulation. Given that waterflooding is the primary method for secondary recovery, understanding its baseline performance is critical for managing these thin-bedded reservoirs. These experiments delineate waterflood front evolution, layer-specific production contributions, and remaining oil distribution patterns under CWF conditions. Identifying the inherent limitations of CWF establishes a necessary baseline for evaluating subsequent enhancement measures, such as unstable injection-production and flow-field reconstruction.

Vertical front evolution was analyzed via cross-sections along the X–Z and Y–Z planes (see Fig. 4a). These cross-sections facilitate the systematic observation of areal sweep dynamics and clarify how permeability contrasts between sub-layers influence water encroachment patterns.

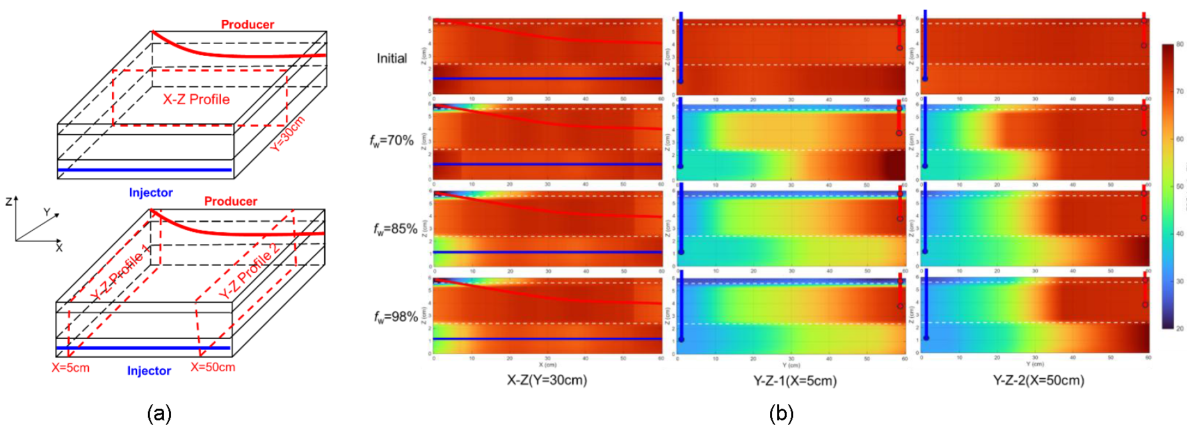

Figure 4—Cross-section orientations and saturation profiles at various locations. (a) Schematic of X–Z and Y–Z cross-section orientations; (b) Saturation profiles along X–Z (Y = 30 cm) and Y–Z (X = 5 cm, X = 50 cm) planes

Waterflooding Front Evolution under Vertical Permeability Contrast. Vertical permeability contrasts trigger pronounced "vertical fingering" in the displacement front (see Fig. 4b). The X–Z cross-section reveals that the high-permeability layer, characterized by low flow resistance, rapidly develops dominant channels that lead to early water breakthrough. Conversely, front advancement in the mid- and low-permeability layers is retarded by interlayer threshold pressure gradients and crossflow effects; as a result, only the mid-perm layer achieves limited sweep by the conclusion of the experiment.

The Y–Z cross-section highlights a severe sweep imbalance. While the high-perm layer reaches a fully water-flooded state, the front in the low-perm layer remains confined near the injector. This disparity causes premature flooding near high-perm production wells (e.g., X = 5 cm), while low-perm zones (e.g., X = 50 cm) remain saturated with unproduced crude oil. Such observations underscore the severe inter-layer interference present even in the early stages of development

Stage-wise Production Succession and Contribution Patterns. Fig. 5 shows the recovery factor and water cut evolution, illustrating both overall performance and layer-specific contributions. From the cumulative recovery factor (CRF) and incremental recovery factor (IRF) curves, three distinct production stages can be identified for the entire model and each individual layer:

1. Stage 1: High oil rate, low water cut stage. Initially, oil production is driven by the high- and mid-permeability layers. However, this stage is brief, as injected water rapidly channels into high-permeability streaks.
2. Stage 2: Rapid water cut increase. Following breakthrough in the high-permeability layer, the water cut exceeds 70%. This layer then becomes a thief zone for ineffective water circulation, exerting a "shielding effect" that stagnates recovery in the mid- and low-permeability layers.
3. Stage 3: Production transition fault. At a terminal water cut of 98%, the final recovery factor is only 27.3%. The mid- and low-permeability layers fail to establish effective displacement pressure gradients, leading to a "production transition fault" where their IRF increases slowly. Consequently, the development pattern degrades into "single-layer water cycling" within the high-permeability zone.

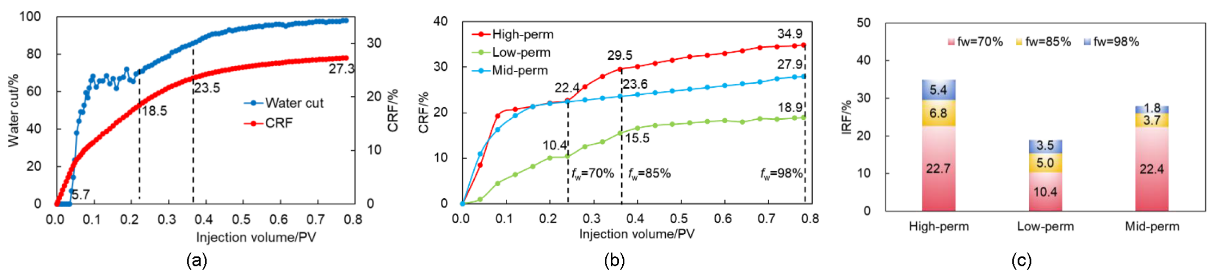

Figure 5—Recovery factor and water cut evolution: overall performance and layer-specific contributions. (a) Overall cumulative recovery factor and water cut curves; (b) Cumulative recovery factor by layer versus injected pore volume; (c) Incremental recovery factor by layer and by production stage

Analysis of Spatial Remaining Oil Distribution and Its Causes. Non-uniform advancement of the displacement front dictates the spatial distribution of remaining oil (Fig. 6). In high-permeability layers, remaining oil is concentrated in poorly swept zones far from the production well. Conversely, insufficient sweep in mid- and low-permeability layers leaves large, contiguous patches of remaining oil between wells, with the low-perm layer exhibiting the highest oil saturation. These results confirm that vertical heterogeneity is the primary geological barrier to mobilizing low-permeability reserves.

The inherent interlayer conflicts that CWF cannot overcome lead to poor sweep efficiency and a production transition fault, leaving substantial reserves stranded in lower-permeability layers—representing the primary production growth opportunity in mature fields. Unlocking this potential requires a staged combination of enhanced measures: unstable waterflooding to mitigate interlayer interference, followed by water shutoff or sidetracking to activate low-permeability zones. The following sections analyze the mechanisms of these three measures within a unified framework, organized by the sequence of their field implementation to maximize production growth.

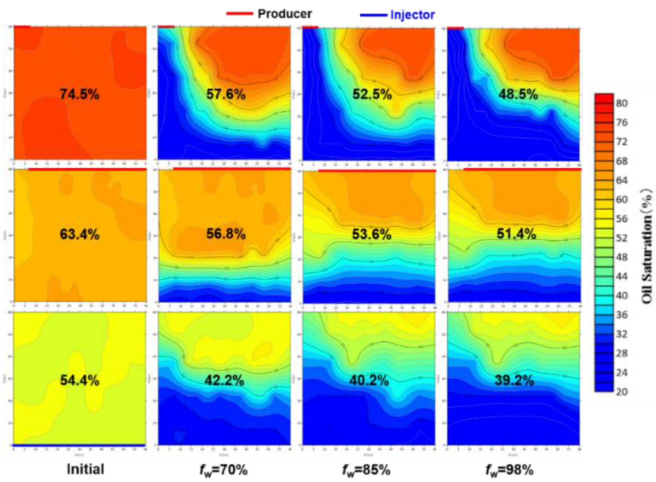

Figure 6—Saturation field in each layer at different stages.

Pressure-Pulse Activation: Mitigating Interlayer Interference via Unstable Waterflooding

Building on the baseline analysis of CWF, this section examines how unstable waterflooding mitigates vertical heterogeneity. In this study, unstable waterflooding is a cyclic injection–production strategy with periodically alternated rates (IPR $\neq$ 1) to generate transient pressure gradients—unlike the steady-state displacement of conventional waterflooding. This approach differs from: Cyclic injection (alternating injection and shut-in periods without adjusting production rates), Pulse injection (short, high-intensity pressure pulses without synchronized production modulation) and Simple rate control (adjusts injection/production rates but maintaining constant IPR over time). The key distinction of our method lies in the synchronized, periodic alternation of both injection and production rates (IPR = 1.2:0.8 and 0.8:1.2 in 40-minute cycles), which creates dynamic pressure waves that propagate through the reservoir and mobilize stagnant oil. Fig. 7 and Fig. 8 show the experimental results between scheme (a) (CWF) and scheme (b) (unstable waterflooding combined with water shutoff).

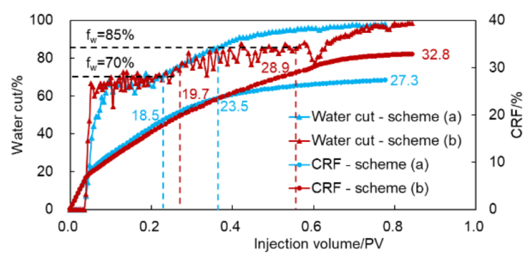

Figure 7—Comparison of dynamic curves of CWF and unstable waterflooding combined with water shutoff

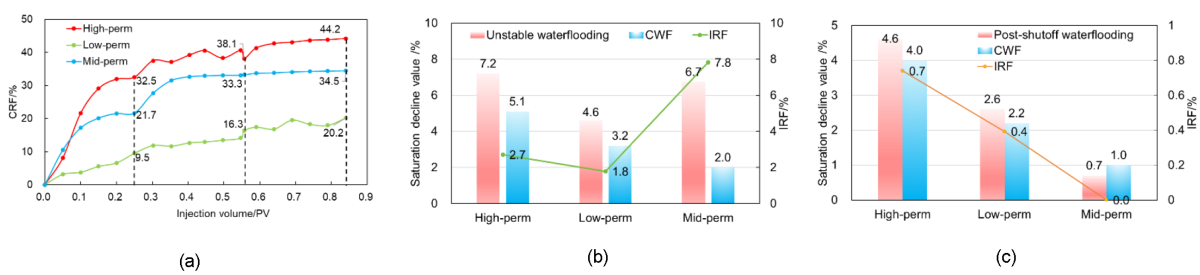

Figure 8—Layer-specific response during unstable waterflooding and post-shutoff stages: recovery contribution and saturation evolution. (a) Cumulative recovery factor by layer versus injected pore volume; (b) Comparison of development effects during unstable waterflooding; (c) Comparison of development effects during post-shutoff waterflooding.

Recovery Response, Water-Cut Control and Layer-Specific Selectivity. Dynamic performance curves reveal marked differences once the water cut exceeds 70%: while CWF exhibits a rapid, monotonic water cut increase, unstable waterflooding maintains fluctuations within a narrow, controlled range. This reflects effective water-cut suppression, substantially extending the production plateau. Unstable waterflooding yielded a 9.2% recovery increase—4.2% higher than CWF—confirming that pressure-pulse activations effectively mobilize stagnant remaining oil.

A detailed layer-specific analysis demonstrates that unstable waterflooding mitigates vertical heterogeneity with distinct selectivity. This differential response is evidenced by the saturation field evolution (Fig. 8a), supported by quantitative data on saturation decline and incremental recovery factor (Fig. 8b):

1. Mid-Permeability Layer (primary contributor). Unstable waterflooding increased oil saturation reduction in this layer from 2.0% (CWF) to 6.7%—a 4.7% incremental gain that represents the primary source of production growth during this stage. Periodic pressure pulses disrupt interlayer shielding, effectively diverting flow into the mid-perm zone. This translates into a 7.8% incremental recovery factor relative to CWF, accounting for the majority of the total 9.2% stage recovery increase. In comparison, the high-perm and low-perm layers contributed only 3.2% and 2.7%, respectively.
2. High- and Low-Permeability Layers (constrained mobilization). While both layers exhibited reduced saturations, their underlying constraints differ, and their incremental recovery factors reflect these limitations. In the high-permeability layer, the severely waterflooding condition reached prior to unstable injection limits any significant further expansion of the swept volume, yielding only a 3.2% incremental recovery factor. In contrast, the low-permeability layer is constrained by a high threshold pressure gradient, which hinders fluid mobilization under the existing pressure-pulse activation, resulting in a marginal 2.7% incremental recovery factor.

Mechanism: Pressure-pulse Activation Induced by Pressure Fluctuations. The key mechanism behind this improvement lies in a fundamental alteration of the pressure field. Unlike the steady-state pressure field characteristic of CWF, unstable waterflooding creates a dynamic pressure environment through periodic adjustments in injection and production rates (Fig. 9). Experimental monitoring confirms that these fluctuations significantly broaden the propagation range of the pressure front, thereby inducing a pronounced pressure-pulse activation within the pore system.

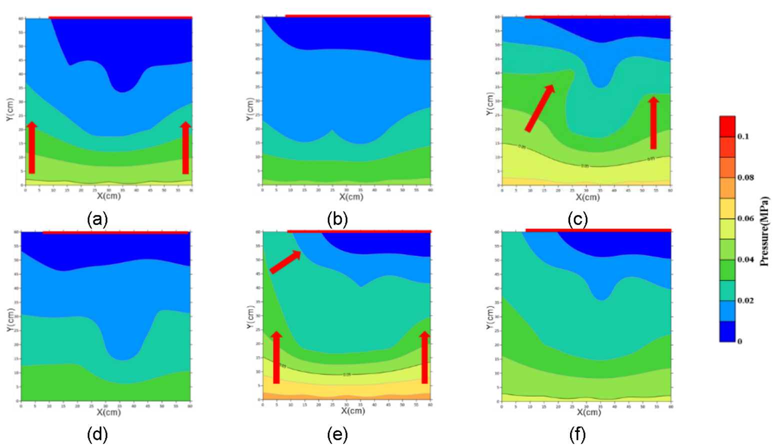

Figure 9—Comparison of pressure fields of CWF and unstable waterflooding combined with water shutoff. (a) Pressure distribution when it reaches the peak stage during continuous waterflooding; (b) Pressure distribution during the stable stage of continuous waterflooding; (c) Pressure distribution under unstable waterflooding with an IPR of 1.2:0.8; (d) Pressure distribution under unstable waterflooding with an IPR of 0.8:1.2; (e) Pressure distribution when it reaches the peak stage; (f) Pressure distribution during the stable stage

To quantitatively describe the difference in pressure variation between unstable waterflooding and CWF, this study defines a Dimensionless Pressure Gradient (*DPG*):

$$DPG = \frac{|P_e - P_s|}{P_s} \tag{1}$$

Where P_e is Reservoir pressure during enhanced waterflooding development (e.g., unstable waterflooding, water shutoff, or sidetracking) at a specific time and location, MPa; P_s is reservoir pressure during CWF (baseline) at the same time and location, MPa.

The *DPG* quantifies the relative intensity of the pressure disturbance induced by unstable waterflooding against the stable waterflooding baseline. By normalizing pressure values at identical spatial coordinates, *DPG* effectively eliminates the influence of absolute pressure levels, enabling a direct and quantitative assessment of both the propagation range and the interference degree of pressure pulses deep within the formation. As shown in Table 4, the *DPG* variation across different Y positions clearly characterizes the spatial distribution and intensity of this pressure disturbance: (1) IPR = 1.2:0.8: System pressure rises rapidly. The pressure gradient near the wellbore increases by 0.3–0.5, while the gradient in deeper formations more than doubles. (2) IPR = 0.8:1.2: The pressure wave propagates forward. As distance increases, the pressure gradient rises further and the pressure decline becomes more pronounced.

Table 4—Dimensionless pressure gradient at different Y positions.

Measures	Stages	Y=10 cm	Y=20 cm	Y=30 cm	Y=40 cm	Y=50 cm
Unstable waterflooding	Injection-production ratio (IPR)=1.2:0.8	0.38	0.50	0.67	1.00	1.25
	Injection-production ratio (IPR)=0.8:1.2	0.38	0.42	0.45	0.60	0.77
Post-shutoff waterflooding	Pressure reaches the peak	0.45	0.60	0.67	1.00	1.13
	Pressure stabilizes	0.18	0.30	0.35	0.54	0.78

Dynamic analysis demonstrates that these periodic pressure fluctuations disrupt the steady-state flow of CWF and significantly enhance the inter-well pressure gradient. This mechanism effectively compensates for the production-transition fault by mobilizing remaining oil in previously stagnant zones of the mid-permeability layer. However, the low-permeability layer remains challenging to activate under the current pressure-fluctuation amplitude, indicating that further synergistic measures—such as sidetracking or plugging—are required to adjust the flow-field distribution and achieve balanced utilization across all layers.

Water Shutoff: Passive Control through Fluid Diversion

Error! Reference source not found. As illustrated in Fig. 8, the development performance before and after water shutoff implementation is also compared. Following water shutoff at the high water-cut stage (f_w > 85%), waterflooding continued following the water shutoff operation. This operation increased stage recovery by 3.9%, resulting in a final recovery factor of 32.8%. This represents a cumulative 5.5% improvement over CWF, accounting for contributions from the preceding unstable waterflooding stage. Specifically, the water shutoff stage itself contributed an incremental 0.4% recovery relative to the end of the unstable waterflooding stage. Layer-specific data show that recovery factors for the high-, mid-, and low-permeability layers increased by 6.1%, 1.2%, and 3.9%, respectively (Fig. 8a).

Saturation analysis (Fig. 8c) indicates that oil saturation decreased by 4.6% in the high-perm layer and 2.6% in the low-perm layer, with mobilization primarily concentrated near the production wellbore. This confirms that the agents successfully plugged high-permeability channels, forcing "fluid diversion".

Consequently, the displacement mechanism transitioned from "dominant channeling" to "balanced multi-layer displacement" as water expanded into previously unswept zones.

Pressure field monitoring after water shutoff (Fig. 9 and Table 4) revealed that, upon reaching peak pressure, the near-wellbore *DPG* increased by 0.4–0.7 times. Once pressure is stabilized, the gradient from the model's middle section to the production area increased by 0.3–0.8 times. This demonstrates that production-well water shutoff effectively blocks dominant flow paths in high-perm layers, significantly increasing flow resistance. Such flow-field adjustment creates the essential conditions for effective fluid diversion.

Sidetracking: Active Mobilization through Path Reconstruction

Compared to water shutoff, sidetracking reconfigures the flow field more thoroughly by relocating the production wellbore. Performance curves comparing scheme (a) and scheme (c) (Fig. 10) show that recovery reached 18.8% at f_w=70% and 28.2% after unstable waterflooding. Subsequent waterflooding after sidetracking to f_w= 98% yielded a final recovery of 35.7%—an 8.4% improvement over CWF.

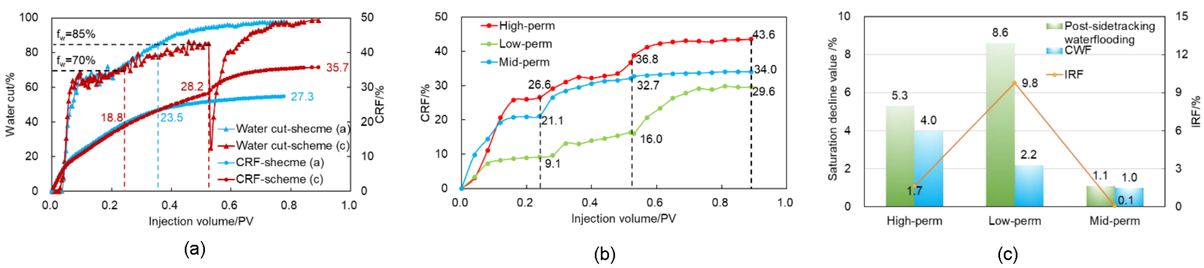

Figure 10—Performance of unstable waterflooding combined with sidetracking. (a) Dynamic curve comparison: CWF vs. unstable flooding + sidetracking; (b) Cumulative recovery factor by layer versus injected pore volume after sidetracking; (c) Comparison of development effects: post-sidetracking waterflooding vs. CWF

Saturation field analysis data reveals that, during post-sidetracking waterflooding stage, CRF increased by 6.8% in the high-perm layer and 13.6% in the low-perm layer, while oil saturation decreased by 5.3% and 8.6%, with corresponding IRF improvements of 1.7% and 9.8%, respectively. These results confirm that sidetracking not only enhances sweep in the already-depleted high-perm layer but, more importantly, unlocks substantial remaining oil in the low-perm layer—accounting for 56% of the total incremental recovery from this scheme. Saturation profile evolution (Fig. 11) reveals that at Y=30 cm, X=5 cm, the mobilization degree of the low-permeability layer significantly increased after sidetracking. However, at X=50 cm, the difference in mobilization of the mid- and low-permeability layers between continuous waterflooding and sidetracking was less pronounced.

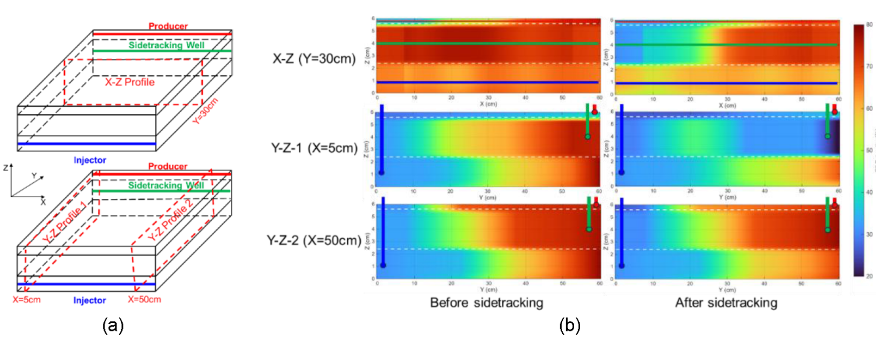

Figure 11—Cross-section orientations and saturation profiles before and after sidetracking. (a) Schematic of X–Z and Y–Z cross-section orientations; (b) Saturation profiles along X–Z (Y = 30 cm) and Y–Z (X = 5 cm, X = 50 cm) planes

The *DPG* at different Y positions (Fig. 12) further shows a significant increase in pressure gradient within approximately one-third of the region from the injection well after sidetracking, confirming a strong path reconstruction effect. This is specifically manifested as: (1) Replacement of dominant flow paths: The original dominant flow path in the high-permeability layer became ineffective due to the relocation of the production point, forcing injected water to establish new displacement channels in the previously unutilized low-permeability zone. (2) Pronounced potential release in low-permeability layer: Saturation field comparison shows that the displacement stream was diverted from the high-permeability to the low-permeability layer, leading to a significant increase in its recovery contribution, accounting for 56% of the total incremental oil from this scheme.

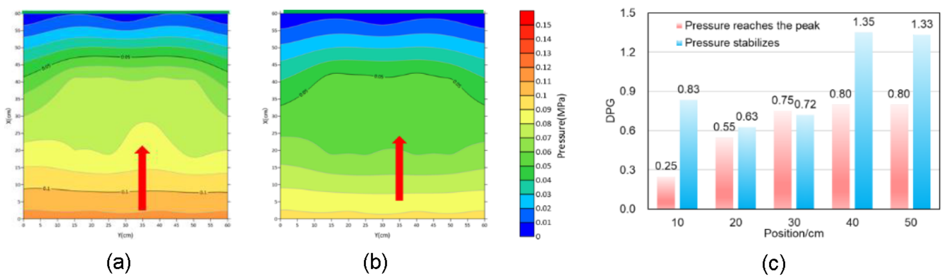

Figure 12—Comparison of pressure fields during subsequent stable waterflooding. (a) Pressure distribution when it reaches the peak stage; (b) Pressure distribution during the stable stage;(c) *DPG* at different Y positions

These results indicate that sidetracking not only alters the spatial path of fluids but also effectively drives the low-permeability thin layer through pressure field redistribution. The mechanistic superiority of sidetracking over water shutoff lies in the fundamental difference between passive diversion and active path reconstruction: Water shutoff operates by blocking existing high-permeability channels, forcing injected water to divert into previously unswept zones. However, this diversion is constrained by the existing wellbore configuration and pressure field; the low-permeability layer still relies on the original injection well to establish displacement, which may not generate sufficient pressure gradient to overcome its threshold. Sidetracking, in contrast, reconfigures the production point itself, relocating the wellbore directly into the low-permeability layer. This fundamentally alters the streamlines: the low-perm layer is no longer a bypassed zone accessed only through crossflow, but becomes the primary target with its own dedicated production conduit. As a result, the displacement front in the low-perm layer advances more efficiently, yielding 56% of incremental oil from this layer alone. In essence, water shutoff improves sweep within the existing flow field, while sidetracking creates an entirely new flow field tailored to access bypassed reserves.

Synergistic Mechanisms: Spatiotemporal Complementarity and Field Implications

Quantitative Synergy Assessment. A comparison of recovery performance across the three schemes (Fig. 13) reveals distinct development outcomes. The CWF baseline exhibits a rapid increase in water cut (f_w > 70%), reaching a final recovery factor of only 27.3%. As see in Fig. 7, significant remaining oil is left in the mid- and low-permeability layers under this scenario. The "unstable flooding + water shutoff" scheme achieved an improved final recovery factor of 32.8%. While this strategy activates mid-perm layers via pressure fluctuations and redirects flow by plugging high-perm channels, its overall sweep efficiency remains limited. The "unstable waterflooding + sidetracking" scheme yielded the optimal results, achieving a recovery factor of 35.7%. This represents an 8.4% improvement over CWF, demonstrating a clear advantage over the water shutoff strategy.

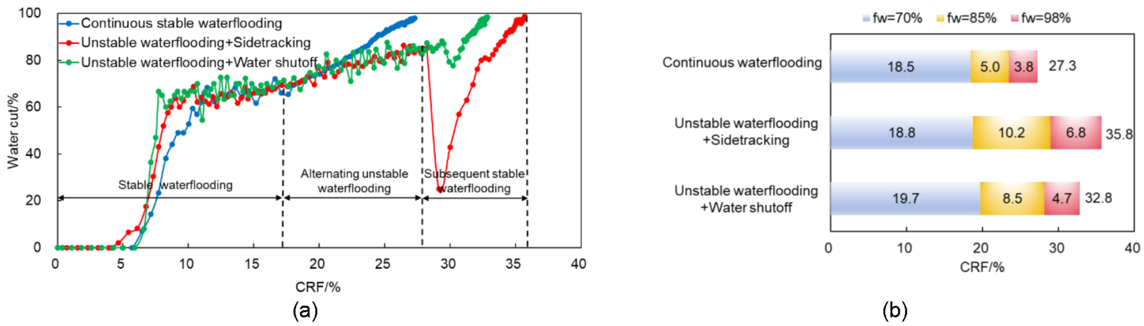

Figure 13—Comparison of development performance for the three schemes. (a) Cumulative oil recovery factor-water cut curves of the three schemes in whole development process; (b) Comparison of EOR effects between different schemes and stages

As illustrated by the saturation field evolution in Fig. 14, this figure enables direct visual comparison of remaining oil distribution and swept efficiency improvements under each intervention strategy. Unstable waterflooding significantly enhances sweep efficiency in the mid-permeability layer, yet its impact on the low-permeability layer is limited. Water shutoff primarily targets the high-permeability layer, mobilizing remaining oil near the production wellbore. In contrast, sidetracking substantially improves sweep efficiency in both high- and low-permeability layers, with the most pronounced mobilization occurring around the sidetracked wellbore. These results confirm that synergistic strategies are significantly superior to single-measure methods for both enhancing oil recovery and suppressing water-cut rise.

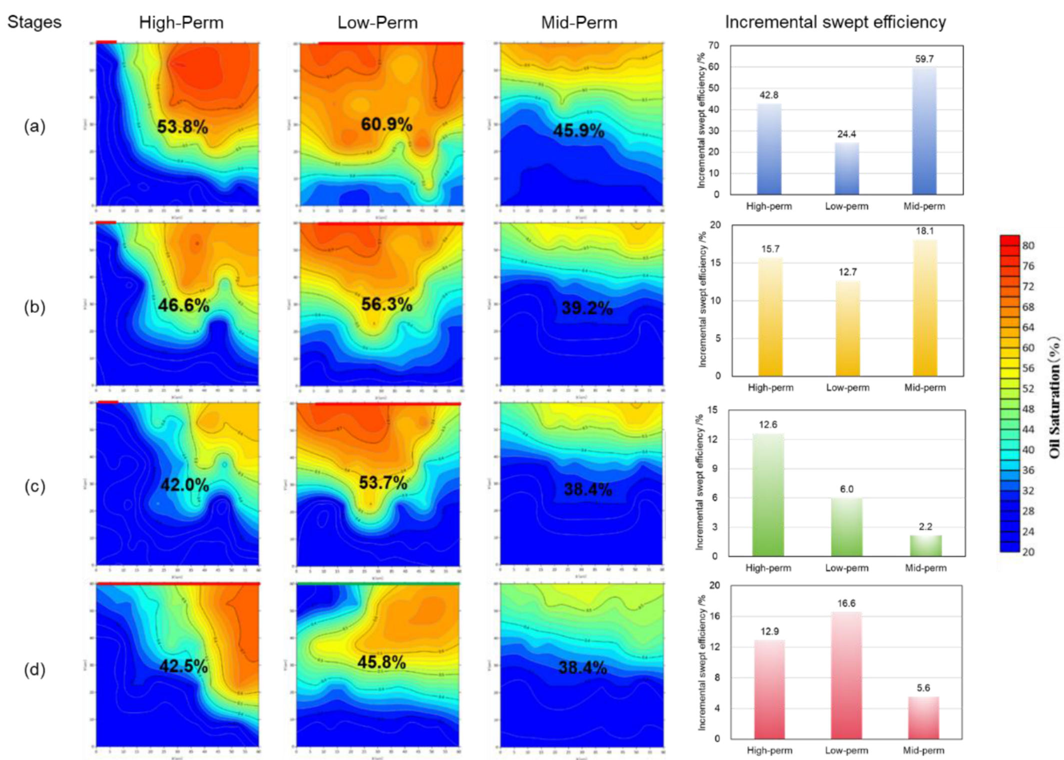

Figure 14—Saturation field evolution across development stages and permeability layers. Each row represents a key development stage: (a) End of CWF; (b) After unstable waterflooding; (c) After water shutoff; (d) After sidetracking. Columns correspond to high-, low-, mid-permeability layers and its incremental swept efficiency from left to right.

Spatiotemporal complementary. The experiments confirm that the synergy of combined measures is non-additive; rather, it represents a systematic spatial-temporal complementarity:

1. Spatial Synergy: Unstable waterflooding activates the mid-perm layer by using pressure fluctuations to overcome threshold pressure gradients. Water shutoff increases flow resistance and redirects fluid by plugging dominant high-perm channels. Sidetracking facilitates "active mobilization" in low-perm layers and unswept zones by establishing new streamlines. Integrating these three measures achieves systematic vertical coverage and balanced mobilization across all permeability tiers.
2. Temporal Succession: Development follows a sequential logic: unstable waterflooding for water-cut stabilization, followed by sidetracking or water shutoff for deep-zone mobilization. During the water cut range of 70%-85%, unstable waterflooding is employed to extend the liquid production plateau. At ultra-high water-cut stages (f_w>85%), field reconfiguration measures are implemented to overcome development bottlenecks.

Integrating experimental data and mechanistic analysis, this study proposes a "Three-Dimensional Synergistic Mobilization Model" for Middle Eastern thin-bedded carbonate reservoirs: (1) Pressure pulse activation: Periodic pressure fluctuations from unstable waterflooding alleviate inter-layer interference, mobilizing stagnant oil in unswept zones. (2) Physical barrier diversion: The physical barriers formed by shutoff agents in dominant channels forcibly divert fluid toward mid- and low-perm layers. (3) Path reconstruction: Sidetracking reconfigures the well pattern, establishing new displacement channels within low-perm layers to improve recovery efficiency in both high- and low-perm zones.

Consequently, the "unstable waterflooding + sidetracking" scheme—characterized by "pressure-pulse activation + path reconstruction"—demonstrates the optimal synergistic effect. It achieves recovery degrees of 43.7% and 29.5% in the high- and low-perm layers, respectively, resulting in a highly balanced development profile. This provides a robust theoretical framework for shifting from "static plugging" to "dynamic reconstruction" when targeting the potential of carbonate reservoirs during ultra-high water-cut stages.

Field-Scale Production Growth Framework. The experimental results translate directly into actionable strategies for production growth in thin-bedded carbonate reservoirs, as summarized in Table 5. Key production growth insights derived from this study include: (1) Threshold-based decision making: The experimental results demonstrate that implementing unstable waterflooding at 70% water cut effectively extends the production plateau. While the quantitative impact of further delay was not systematically investigated in this study, the observed decline in mid-permeability layer responsiveness suggests that timely intervention is critical—a hypothesis warranting further investigation through dedicated experimental design. (2) Layer-specific intervention timing: Mid-permeability layers respond best to pressure pulsation alone, while low-permeability layers require sidetracking to overcome threshold pressure gradients—a distinction critical for allocating workover resources. (3) Synergistic gain quantification: The "unstable flooding + sidetracking" combination delivers 8.4% absolute recovery gain over continuous flooding, with 56% of incremental oil contributed by the low-permeability layer—demonstrating that true production growth in mature fields requires targeted activation of bypassed zones, not merely improved sweep in already-drained layers. (4) Economic limit extension: The optimal combination delays the economic limit (f_w=98%) by an estimated 2-3 years in analogous field applications, based on production decline curve analysis.

The "unstable flooding + sidetracking" scheme delivers the highest incremental recovery, but its field application must be guided by economic and operational realities. Advances in horizontal drilling have reduced costs and improved recovery, with sidetracked wells typically producing 3–5 times more than adjacent vertical wells (Zhang et al. 2019, Yuan et al. 2020, Emelander 2022). However, its higher upfront cost—rig mobilization, wellbore preparation, and downtime—limits its use to reservoirs with significant remaining oil potential in low-perm zones. Candidate wells should meet three criteria: (1)

confirmed bypassed pay (via production logging or saturation monitoring); (2) mechanical integrity for lateral completion; (3) target layer continuity sufficient to justify investment.

Table 5—Field implementation framework for production growth

Development Stage	Water Cut Threshold	Optimal Measure	Primary Mechanism	Expected Production Impact
Early stage	< 70%	Continuous waterflooding	Baseline development	Establish production plateau
Mid-high stage	70% – 85%	Unstable waterflooding	Pressure pulse activation	Extend plateau, reduce water cut rise rate
High water cut stage	85% – 90%	Water shutoff	Passive diversion (block thief zones)	3-5% incremental recovery, mobilizes near-wellbore remaining oil in high-perm
Ultra-high water cut stage	> 90%	Sidetracking	Path reconstruction, low-perm layer activation	8-10% incremental recovery, balanced multi-layer production

Water shutoff remains the preferred low-cost option for immediate water cut control in thief zones. The optimal field strategy is therefore staged: water shutoff for quick response in high-perm zones, and sidetracking for long-term production growth from low-perm layers. Regarding application timing, sidetracking is generally not recommended earlier in field life (f_w<70%) for two reasons: (1) its higher cost is not justified when simpler methods suffice; and (2) its value lies in accessing low-perm layers unswept by conventional flooding—a bottleneck that only emerges at late stages. It is therefore best reserved for f_w>85% when low-perm layer activation becomes the primary production growth challenge.

The experimental results presented in this study are directly applicable to thin-bedded carbonate reservoirs characterized by: Strong vertical heterogeneity with permeability contrasts exceeding 10:1 between layers; Presence of thief zones at the reservoir top that cause premature water breakthrough; Well-developed interlayers that impede vertical crossflow; Remaining oil saturation >40% in mid- to low-permeability layers after conventional waterflooding.

The applicability of each enhanced measure is governed by reservoir conditions and production stage: Unstable waterflooding is most effective when water cut ranges from 70% to 85%. Below 70%, the benefit of pressure pulsing is marginal; above 85%, the low-permeability layer remains largely unresponsive to pressure-pulse activations alone. Water shutoff is recommended for wells with confirmed high-permeability thief zones and water cut exceeding 85%, where rapid water cycling dominates production. Sidetracking is suitable for wells with confirmed bypassed pay in low-permeability layers (oil saturation >50%), mechanical integrity for lateral completion, and sufficient layer thickness (>2 m) to justify the investment.

Regarding applicability to extremely tight layers (permeability <1 mD): The threshold pressure gradient in such layers typically exceeds the pressure pulse amplitude achievable by unstable waterflooding alone. For these reservoirs, sidetracking—or even more aggressive stimulation measures (e.g., acid fracturing)—would be required to establish effective displacement.

These findings provide reservoir engineers with a quantitative, stage-gated decision matrix for selecting and sequencing EOR measures to achieve sustained production growth throughout field life.

Conclusions

Through large-scale 3D physical simulations, this study systematically characterizes the synergistic mechanisms of unstable waterflooding, water shutoff, and sidetracking in Middle Eastern thin-bedded carbonate reservoirs. The primary conclusions are as follows:

1. CWF limitations: CWF is physically constrained by extreme vertical heterogeneity. Injected water preferentially breaks through high-permeability layers, leading to ineffective water cycling and

a "production transition fault." This results in a low ultimate recovery factor of 27.3%, leaving substantial reserves unswept in mid-to-low permeability zones—representing the production growth potential to be unlocked by synergistic measures.

2. Pressure pulse activation: Unstable waterflooding improves interlayer utilization through pressure field reconstruction. Dimensionless Pressure Gradient (*DPG*) analysis confirms that periodic perturbations more than double the effective inter-well pressure gradient. This effectively overcomes the threshold pressure gradients of mid-permeability layers, yielding a 7.8% stage recovery increase and suppressing rapid water-cut rise. This extends the production plateau and reduces water cut rise rate, directly contributing to sustained field production growth during the mid-development stage.
3. Flow field reconfiguration: Water shutoff and sidetracking play complementary roles during ultra-high water-cut stages. Water shutoff serves as a "passive control" by plugging high-perm thief zones. This measure restores production from watered-out intervals, adding incremental barrels with minimal capital expenditure. Conversely, sidetracking enables "active mobilization" by re-establishing streamlines to target low-perm zones, delivers the most significant production growth impact, contributing 8.4% incremental recovery with 56% of incremental oil from previously unproductive low-permeability layers.
4. Optimal synergy: The "unstable waterflooding + sidetracking" combination demonstrates the highest synergistic efficiency, achieving an ultimate recovery factor of 35.7%, demonstrating that maximizing production growth in mature carbonate fields requires a stage-gated, synergistic approach that systematically activates all permeability layers. This synergy is defined by spatial-temporal complementarity: unstable waterflooding first activates mid-perm layers via pressure fluctuations, while sidetracking subsequently facilitates deep mobilization of low-perm zones.
5. Strategic framework: We propose a field-scale development sequence for production growth: Fluctuation driven activation – Flow field reconstruction – Balanced displacement. This targeted model provides a robust theoretical framework for shifting from "static plugging" to "dynamic system reconstruction" in heterogeneous carbonate reservoirs during high water-cut stages.

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